

# Calculus: *Limits and Sequences*

Autumn Term 2022

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**First Part of Assessed Coursework:** Questions **1, 2(b), 3, 6(c)** are assessed and are due by **7pm on Tuesday 26 October**.

1. What is the *least upper bound* or supremum and *greatest lower bound* or infimum in the reals  $\mathbb{R}$  of the following sets of numbers:

- (a)  $\{x \in \mathbb{N} \mid 1 \leq x^2 \leq 29\}$
- (b)  $\{x \in \mathbb{Q} \mid 1 \leq x^2 \leq 29\}$
- (c)  $\{x \in \mathbb{R} \mid 1 \leq x^2 \leq 29\}$

In each case, state whether the infimum and supremum are in the given set.

2. For each of the sequences below  $(a_n)_{n \geq 1}$ , guess the limit and prove directly<sup>1</sup> that the sequence tends to that limit.

- (a)  $a_n = -\frac{1}{\sqrt{2n}}$  for all  $n \geq 1$
- (b)  $a_n = \frac{1 - e^{-n}}{2}$  for all  $n \geq 1$
- (c)  $a_n = \frac{n-1}{n}$  for all  $n \geq 1$
- (d)  $a_n = C$  for all  $n \geq 1$ , where  $C$  is a real constant.

3. Use the Sandwich theorem and any previous result to prove that:

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n^2}\right) = 1$$

4. **Exam standard.** You are given that both of the sequences  $(a_n)_{n \geq 1}$ , with  $a_n$  defined for  $n \geq 1$  below, tend to 0. Prove that they do so directly or using the Sandwich theorem.

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<sup>1</sup>Direct proof of sequence convergence requires that you use the  $\epsilon$ - $N$  method.

$$(a) \ a_n = \begin{cases} 1/n & : \text{if } n < 1000 \\ 1/n^2 & : \text{if } n \geq 1000 \end{cases}$$

$$(b) \ a_n = \begin{cases} 1/n & : \text{if } n \text{ even} \\ -1/n & : \text{if } n \text{ odd} \end{cases}$$

5. The sequence  $(a_n = 2^{-n})_{n \geq 1}$  converges to 0. Attempt a direct proof that  $\lim_{n \rightarrow \infty} a_n = 1/8$  instead and explain where it fails.

6. **Exam standard.**

- (a) Let  $\alpha$  be a positive, real constant. Use a direct proof to show that the sequence  $(a_n)_{n \geq 1} = (n^{-\alpha})_{n \geq 1}$  converges to 0 as  $n$  tends to infinity.  
 (b) Use the Sandwich theorem to show that

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

- (c) Let  $p$  be a fixed natural number. What happens to the sequence

$$\left(\frac{n!}{n^p}\right)_{n \geq 1}$$

as  $n \rightarrow \infty$ ?

7. **Exam standard.** Let  $\theta$  be any constant in  $\mathbb{R}$ . Using any techniques of your choice, prove that the following sequence converges to 0 as  $n \rightarrow \infty$ :

$$(a_n)_{n \geq 1} = \left(\frac{\sin n\theta}{2^n}\right)_{n \geq 1}$$

8. Let  $a_n = \left(1 + \frac{1}{n}\right)^n$  for all  $n \geq 1$ . Show that

- (a)  $(a_n)_{n \geq 1}$  is an increasing sequence.  
 (b)  $(a_n)_{n \geq 1}$  is upper bounded by  $e$ .

*Hint:* You may use the fact that  $\ln x \leq x - 1$ .

9. Let  $a_1 = 1$  and define  $a_{n+1} = \sqrt{1 + 2a_n}$  for all  $n \geq 1$ .

- (a) Show that the sequence  $(a_n)_{n \geq 1}$  is increasing.  
 (b) Show that the sequence  $(a_n)_{n \geq 1}$  is upper bounded by 3.  
 (c) Explain why  $\lim_{n \rightarrow \infty} a_n$  exists and determine its value.